

GRAND FORKS INTL, ND, USA

WMO: 727570

Lat: 47.943N

Lon: 97.184W

Elev: 257

StdP: 98.28

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14916

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.8	-27.1	-33.1	0.2	-29.4	-30.5	0.2	-26.9	14.5	-15.4	12.9	-13.4	3.3	300	0.681

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	31.6	22.2	30.0	21.0	28.5	20.1	24.0	29.3	22.7	28.0	21.5	26.6	5.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	17.4	27.6	20.9	16.0	26.1	19.6	14.8	24.6	73.3	29.4	68.2	28.0	63.7	26.4	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.3	11.0	9.5	DB	-33.5	34.8	3.9	1.3	-36.2	35.7	-38.5	36.5	-40.7	37.2	-43.5	38.1	(4)
(5)			WB	-33.1	26.0	3.3	1.1	-35.5	26.8	-37.4	27.5	-39.3	28.1	-41.7	28.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.8	-14.0	-11.8	-4.0	5.2	12.5	18.5	20.9	19.8	14.9	6.8	-2.4	-10.4
	DBStd	13.66	8.12	7.69	7.54	5.64	4.92	3.81	3.06	3.25	4.69	5.11	6.39	7.26
	HDD10.0	3123	743	611	437	160	31	1	0	0	11	126	374	631
	HDD18.3	5192	1001	844	694	394	189	43	12	22	122	358	623	889
	CDD10.0	1211	0	0	1	16	108	255	339	304	159	27	1	0
	CDD18.3	237	0	0	0	0	8	48	92	68	20	1	0	0
	CDH23.3	2588	0	0	0	8	144	518	936	719	243	19	0	0
	CDH26.7	786	0	0	0	1	41	153	291	225	72	3	0	0
Wind	WSAvg	4.7	4.9	4.9	5.0	5.1	5.0	4.4	3.8	3.9	4.4	4.8	4.8	4.8
Precipitation	PrecAvg	493	16	12	22	34	59	81	76	67	52	35	20	15
	PrecMax	641	45	42	78	92	146	200	190	158	142	147	100	50
	PrecMin	320	2	1	1	0	4	17	12	9	7	2	0	2
	PrecStd	88	13	11	16	25	35	47	47	34	35	34	20	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.5	6.8	17.2	24.1	30.9	33.5	33.1	33.5	31.7	26.1	15.8	5.5
		MCWB	2.4	3.8	11.5	13.3	18.8	22.5	24.0	23.5	21.5	15.9	10.1	2.4
	2%	DB	2.2	3.5	11.8	20.8	27.3	30.1	31.3	31.1	28.8	21.4	11.8	2.7
		MCWB	0.7	1.7	8.1	11.9	16.4	20.4	23.0	21.6	19.5	14.2	7.5	1.0
	5%	DB	0.0	1.6	8.1	17.7	24.2	28.3	29.7	29.1	26.3	17.8	8.8	1.0
		MCWB	-1.1	0.5	5.4	10.4	14.8	19.3	22.0	20.8	18.1	12.3	5.6	-0.2
	10%	DB	-2.5	-0.4	5.0	14.9	21.9	26.4	28.1	27.4	23.5	15.1	6.0	-0.7
		MCWB	-3.4	-1.5	3.3	9.0	13.8	18.6	20.8	19.7	16.8	10.6	3.5	-1.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.5	4.3	12.8	14.8	20.2	24.8	25.8	25.0	23.0	17.8	10.8	3.0
		MCDB	4.3	6.6	15.8	21.8	27.5	30.2	31.0	31.2	29.6	23.5	14.0	5.1
	2%	WB	0.8	1.9	8.5	12.4	18.2	22.5	24.3	23.2	20.8	15.2	8.1	1.1
		MCDB	2.0	3.1	11.6	19.1	24.1	27.8	29.3	29.0	26.2	20.0	11.4	2.5
	5%	WB	-1.1	0.5	5.4	10.8	16.4	20.8	23.1	21.9	19.2	12.9	5.7	-0.1
		MCDB	0.0	1.6	7.9	16.8	22.2	26.1	28.1	27.2	23.9	17.4	8.5	1.0
	10%	WB	-3.3	-1.5	3.2	9.4	15.0	19.5	21.9	20.7	17.8	10.7	3.6	-1.6
		MCDB	-2.5	-0.5	5.0	14.6	20.2	24.3	26.8	25.7	22.5	14.5	5.8	-0.7
Mean Daily Temperature Range	5% DB	MDBR	9.4	10.1	9.4	11.5	12.9	11.9	12.3	13.1	12.8	10.7	9.1	8.6
		MCDBR	10.2	9.9	12.6	17.2	16.8	14.7	14.1	15.1	16.4	15.9	13.3	10.2
		MCWBR	9.1	8.4	9.3	10.0	8.5	7.1	7.3	7.3	8.3	9.4	9.3	8.7
	5% WB	MCDBR	10.2	9.4	12.2	15.7	13.7	12.4	12.7	13.3	13.8	14.7	12.6	9.8
		MCWBR	9.2	8.0	9.2	9.5	7.8	6.9	7.4	7.3	8.0	9.2	9.0	8.5
Clear-Sky Solar Irradiance	taub	0.283	0.301	0.301	0.353	0.369	0.383	0.383	0.384	0.349	0.306	0.277	0.265	
	taud	2.286	2.258	2.440	2.356	2.373	2.362	2.365	2.363	2.454	2.545	2.517	2.414	
	Ebn at Noon	786	853	925	899	894	880	874	856	858	853	812	779	
	Edh at Noon	74	95	94	114	116	119	117	112	93	72	59	58	
All-Sky Solar Radiation	RadAvg	1.58	2.78	3.91	4.87	5.49	5.99	6.34	5.29	3.88	2.40	1.42	1.09	
	RadStd	0.17	0.25	0.41	0.44	0.35	0.45	0.27	0.36	0.27	0.35	0.17	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page